Phys 115B HW 3

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1 ND

Question 1. *Hydrogen-like (variation on 4.19, or 4.16 in second edition [or 4.17 in the first edition]) Energy levels and eigenfunctions of a "hydrogenic" system can be obtained from those of the hydrogen atom by appropriately scaling mass and charge. Tritium (^3H) and $^3He^+$ are examples, which are linked by the process of beta decay, whereby a neutron in the tritium nucleus can convert to a proton by emitting a neutrino and an energetic electron which escapes the resulting $^3He^+$ ion.*

(a) *For a 1-electron atom with a nucleus of charge Ze, find an expression for the Bohr radius and the Bohr energies as appropriate multiples of the hydrogen values*

(b) *Find the Bohr radius of tritium (you can approximate m_e/m_p as zero)*

(c) *Find the Bohr radius of $^3He^+$*

(d) *A tritium atom is in its ground state. It undergoes a beta decay, which turns the tritium nucleus into a ^3He nucleus. Assume that the decay is instantaneous and that it does not disturb the wavefunction of the bound electron. Calculate the probability that the electron will be found in the ground state of $^3He^+$*

(e) *The emitted electron leaves with an energy of about 6 keV. The recoil from this emission gives a kick to the ^3He nucleus (Newton's second law and all that). Find the energy and velocity of the ^3He recoil, and determine whether we were correct to ignore this effect in the previous part.*

Proof. □

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Question 2. *Commutators*

(a) *Work out the commutators* $[L_z, x]$, $[L_z, p_x]$, $[L_z, y]$, $[L_z, p_y]$, $[L_z, z]$, $[L_z, p_z]$.

(b) *Use those results to verify that* $[L_z, L_x] = i\hbar L_y$

(c) *Find the commutators* $[L_z, r^2]$ *and* $[L_z, p^2]$.

(d) *Show that a Hamiltonian with a spherically symmetric potential* $V(r) = V(r)$ *commutes with all components of* $\mathbf{L}$, *and that* H, L^2, *and* L_z *are compatible observables.*

Proof. First, recall the definitions of each angular momentum operators:

$$L_x = yp_z - zp_y, \quad L_y = zp_x - xp_z, \quad L_z = xp_y - yp_x \tag{1}$$

Also, recall the commutation relations of position and momentum operators:

$$u, v \in \{x, y, z\}, \quad [u, p_v] = i\hbar\delta_{uv}, \quad [u, v] = 0, \quad [p_u, p_v] = 0 \tag{2}$$

(a) The desired commutators are as follow:

$$\begin{aligned}
[L_z, x] &= (xp_y - yp_x)x - x(xp_y - yp_x) && (3)\\
&= x^2p_y - y(xp_x - i\hbar) - x^2p_y + xyp_x && (4)\\
&= i\hbar y && (5)
\end{aligned}$$

$$\begin{aligned}
[L_z, p_x] &= (xp_y - yp_x)p_x - p_x(xp_y - yp_x) && (6)\\
&= xp_yp_x - y(p_x)^2 - (xp_x - i\hbar)p_y + y(p_x)^2 && (7)\\
&= i\hbar p_y && (8)
\end{aligned}$$

$$\begin{aligned}
[L_z, y] &= (xp_y - yp_x)y - y(xp_y - yp_x) && (9)\\
&= x(yp_y - i\hbar) - y^2p_x - yxp_y + y^2p_x && (10)\\
&= -i\hbar x && (11)
\end{aligned}$$

$$\begin{aligned}
[L_z, p_y] &= (xp_y - yp_x)p_y - p_y(xp_y - yp_x) && (12)\\
&= x(p_y)^2 - yp_xp_y - x(p_y)^2 + (yp_y - i\hbar)p_x && (13)\\
&= -i\hbar p_x && (14)
\end{aligned}$$

$$[L_z, z] = (xp_y - yp_x)z - z(xp_y - yp_x) = 0 \tag{15}$$

$$[L_z, p_z] = (xp_y - yp_x)p_z - p_z(xp_y - yp_x) = 0 \tag{16}$$

(b) Using the commutation relation from part (a), we have:

$$[L_z, L_x] = L_z(yp_z - zp_y) - (yp_z - zp_y)L_z \tag{17}$$
$$= (yL_z - i\hbar x)p_z - zL_zp_y - (yp_z - zp_y)L_z \tag{18}$$
$$= yp_zL_z - i\hbar xp_z - z(p_yL_z - i\hbar p_x) - (yp_z - zp_y)L_z \tag{19}$$
$$= i\hbar(zp_x - xp_z) = i\hbar L_y \tag{20}$$

(c) Again, using the commutation relations in (a), we have:

$$[L_z, r^2] = [L_z, x^2] + [L_z, y^2] + [L_z, z^2] \tag{21}$$
$$= (L_zx)x - x(xL_z) + (L_zy)y - y(yL_z) \tag{22}$$
$$= (L_zx)x - (xL_z)x + x(L_zx) - x(xL_z) + (L_zy)y - (yL_z)y + y(L_zy) - y(yL_z) \tag{23}$$
$$= [L_z, x]x + x[L_z, x] + [L_z, y]y + y[L_z, y] \tag{24}$$
$$= i\hbar yx + x \cdot i\hbar y - i\hbar xy + y \cdot (-i\hbar x) \tag{25}$$
$$= 0 \tag{26}$$

$$[L_z, p^2] = [L_z, p_x^2] + [L_z, p_y^2] + [L_z, p_z^2] \tag{27}$$
$$= (L_zp_x)p_x - p_x(p_xL_z) + (L_zp_y)p_y - p_y(p_yL_z) \tag{28}$$
$$= (L_zp_x)p_x - (p_xL_z)p_x + p_x(L_zp_x) - p_x(p_xL_z) + (L_zp_y)p_y - (p_yL_z)p_y + p_y(L_zp_y) - p_y(p_yL_z) \tag{29}$$
$$= [L_z, p_x]p_x + p_x[L_z, p_x] + [L_z, p_y]p_y + p_y[L_z, p_y] \tag{30}$$
$$= i\hbar p_yp_x + p_x \cdot i\hbar p_y - i\hbar p_xp_y + p_y \cdot (-i\hbar p_x) \tag{31}$$
$$= 0 \tag{32}$$

(d) Consider a Hamiltonian with spherically symmetric potential $H = \frac{p^2}{2m} + V(r)$, first it's clear that from part (c), one has $[L_z, p^2] = 0$. After changing the coordinates (i.e. swapping order of x, y, z), using the spatial symmetry one can also claim that $[L_u, p^2] = 0$ for all $u \in \{x, y, z\}$.

So, to prove each component of $\mathbf{L}$ commutes with H, it suffices to consider for the spherically symmetric potential $V(r)$.

Given the definition $r = \sqrt{x^2 + y^2 + z^2}$, one has $\frac{\partial f}{\partial u} = \frac{\partial r}{\partial u}\frac{\partial f}{\partial r} = \frac{u}{\sqrt{x^2+y^2+z^2}}\frac{\partial f}{\partial r} = \frac{u}{r}\frac{\partial f}{\partial r}$ for any differentiable function f. Therefore, we have:

$$[L_z, V(r)] = (xp_y - yp_x)V(r) - V(r)(xp_y - yp_x) \tag{33}$$
$$= -i\hbar\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right)V(r) - V(r)(xp_y - yp_x) \tag{34}$$
$$= -i\hbar\left(x\left(\frac{y}{r}\frac{\partial V}{\partial r} + V(r)\frac{\partial}{\partial y} - y\left(\frac{x}{r}\frac{\partial V}{\partial r} + V(r)\frac{\partial}{\partial x}\right)\right)\right) - V(r)(xp_y - yp_x) \tag{35}$$
$$= -i\hbar \cdot V(r)(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}) - V(r)(xp_y - yp_x) \tag{36}$$
$$= V(r)(xp_y - yp_x) - V(r)(xp_y - yp_x) = 0 \tag{37}$$

Hence, this shows that L_z commutes with $V(r)$ as operator. Since L_z commutes with all the components of the Hamiltonian H, one can say $[L_z, H] = 0$. By the same spatial symmetry argument, one can also show that $[L_u, H] = 0$ for all $u \in \{x, y, z\}$. This concludes that all components of $\mathbf{L}$ commutes with H.

Finally, since $L^2 = L_x^2 + L_y^2 + L_z^2$, it's clear that all of its components commute with H (when H is endowed with a spherically symmetric potential). Also, from the calculation before it's clear that L_z commutes with H, and L^2 commutes with L_z. Therefore, H, L^2, L_z are compatible observables.

$\square$

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Question 3. *Normalizing L_+ and L_- (Griffiths 4.21, 4.18 in second edition)*
The raising and lowering operators change the value of m by one unit:

$$L_+ f_\ell^m = (A_\ell^m) f_\ell^{m+1}, \quad L_- f_\ell^m = (B_\ell^m) f_\ell^{m-1}$$

where A_ℓ^m and B_ℓ^m are constants.
Questions: *What* are *they, if the eigenfunctions are to be normalized?*
Hint: *First show that $L_\mp$ is the hermitian conjugate of $L_\pm$ (since L_x and L_y are observables, you may assume they are hermition...but prove it if you like); then use the following equation:*

$$L^2 = L_\pm L_\mp + L_z^2 \mp \hbar L_z$$

Answer:

$$A_\ell^m = \hbar\sqrt{\ell(\ell+1) - m(m+1)} = \hbar\sqrt{(\ell-m)(\ell+m+1)}$$
$$B_\ell^m = \hbar\sqrt{\ell(\ell+1) - m(m-1)} = \hbar\sqrt{(\ell+m)(\ell-m+1)}$$

Note what happens at the top and bottom of the ladder (i.e. when you apply L_+ to f_ℓ^ℓ or L_- to $f_\ell^{-\ell}$)

Proof. First, recall the definition $L_\pm := L_x \pm iL_y$. Hence, its adjoint (Hermitian Conjugate) can be given as follow:

$$L_\pm^* = L_x^* \pm (iL_y)^* = L_x \pm (-iL_y) = L_x \mp i_y = L_\mp \tag{38}$$

Now, recall that $L^2 f_\ell^m = \hbar^2 \ell(\ell+1) f_\ell^m$ and $L_z f_\ell^m = \hbar m f_\ell^m$. So, one gets the following:

$$\hbar^2 \ell(\ell+1) f_\ell^m = L^2 f_\ell^m = (L_+ L_- + L_z^2 - \hbar L_z) f_\ell^m = L_+ L_- f_\ell^m + \hbar^2 m(m-1) f_\ell^m \tag{39}$$

$$\hbar^2 \ell(\ell+1) f_\ell^m = L^2 f_\ell^m = (L_- L_+ + L_z^2 + \hbar L_z) f_\ell^m = L_- L_+ f_\ell^m + \hbar^2 m(m+1) f_\ell^m \tag{40}$$

Which implies the following two equality:

$$\hbar^2(\ell(\ell+1) - m(m-1)) f_\ell^m = L_+ L_- f_\ell^m, \quad \hbar^2(\ell(\ell+1) - m(m+1)) f_\ell^m = L_- L_+ f_\ell^m \tag{41}$$

As a result, one derives the following:

$$|A_\ell^m|^2 = \left\langle A_\ell^{m+1} f_\ell^m \middle| A_\ell^{m+1} f_\ell^m \right\rangle = \langle L_+ f_\ell^m | L_+ f_\ell^m \rangle = \langle f_\ell^m | L_- L_+ f_\ell^m \rangle = \hbar^2(\ell(\ell+1) - m(m+1)) \tag{42}$$

$$|B_\ell^m|^2 = \left\langle B_\ell^m f_\ell^{m-1} \middle| B_\ell^m f_\ell^{m-1} \right\rangle = \langle L_- f_\ell^m | L_- f_\ell^m \rangle = \langle f_\ell^m | L_+ L_- f_\ell^m \rangle = \hbar^2(\ell(\ell+1) - m(m-1)) \tag{43}$$

Hence, take the value to be nonnegative real, one gets $A_\ell^m = \hbar\sqrt{\ell(\ell+1) - m(m-1)}$, and $B_\ell^m = \hbar\sqrt{\ell(\ell+1) - m(m+1)}$. □

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Question 4. *Rigid Rotor (Griffiths 4.27 a, b, c only, 4.24 in the second edition)*
Two particles (masses m_1 and m_2) are attached to the ends of a massless rigid rod of length a. The system is free to rotate in three dimensions about the (fixed) center of mass.

(a) *Show that the allowed energies of this* ***rigid rotor*** *are*

$$E_n = \frac{\hbar^2}{2I} n(n+1) \quad (n = 0, 1, 2, ...), \qquad I := \frac{m_1 m_2}{m_1 + m_2} a^2$$

is the moment of inertia of the system.

Hint: *First express the (classical) energy in terms of the angular momentum.*

(b) *What are the normalized eigenfunctions for this system? (Let θ and ϕ define the orientation of the rotor axis.) What is the degeneracy of the nth energy level?*

(c) *What spectrum would you expect for this system? (Give a formula for the frequencies of the spectral lines.)*

Answer: $\nu_j = \frac{\hbar j}{2\pi I}$, $j = 1, 2, 3, ...$

(d) *Figure 4.13 shows a portion of the rotational spectrum of carbon monoxide (CO). What is the frequency separation ($\Delta\nu$) between adjacent lines? Look up the masses of ^{12}C and ^{16}O, and from m_1, m_2, and $\Delta\nu$ determine the distance between the atoms.*

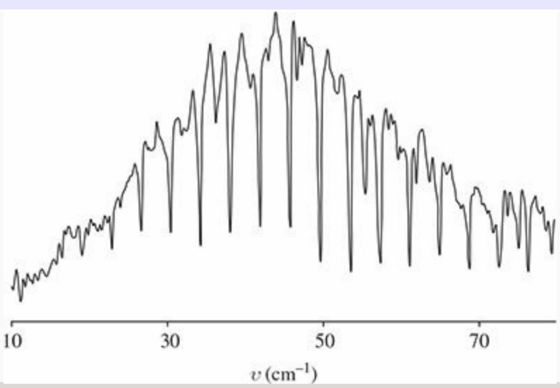

Figure 4.13: Rotation spectrum of CO. Note that the frequencies are in spectroscopist's units: inverse centimeters. To convert to Hertz, multiply by $c = 3.00 \times 10^{10}$ cm/s. Reproduced by permission from John M. Brown and Allan Carrington, *Rotational Spectroscopy of Diatomic Molecules*, Cambridge University Press, 2003, which in turn was adapted from E. V. Loewenstein, *Journal of the Optical Society of America*, **50**, 1163 (1960).

Proof.

(a) First, recall that the classical energy of the system (which is assumed to have no potential) is given by $E = T = \frac{L^2}{2I}$, where L indicates the (classical) total angular momentum's magnitude, while I represents the moment of inertia of the system. So, the hamiltonian $H = \frac{L^2}{2I}$.

Then, with the rod of length a, the center of mass would located at $\frac{m_2 a}{m_1+m_2}$ away from particle m_1, while located at $\frac{m_1 a}{m_1+m_2}$ away from particle m_2. Then, the total moment of inertia is given as follow:

$$I = I_1 + I_2 = m_1\left(\frac{m_2 a}{m_1+m_2}\right)^2 + m_2\left(\frac{m_1 a}{m_1+m_2}\right)^2 \tag{44}$$

$$= \frac{m_1+m_2}{(m_1+m_2)^2}m_1 m_2 a^2 = \frac{m_1 m_2}{m_1+m_2}a^2 \tag{45}$$

Now, given any eigenstate of this Hamiltonian (which is a scaling of the total angular momentum operator L^2), it's in the form of f_ℓ^m with $\ell \in \mathbb{N}$, and $m \in \{-\ell, -\ell+1, ..., \ell-1, \ell\}$, where $L^2 f_\ell^m = \hbar^2\ell(\ell+1)$. Hence, the corresponding eigenvalue is as follow:

$$E_\ell = \left\langle f_\ell^m \middle| \frac{L^2}{2I} f_\ell^m \right\rangle = \frac{\hbar^2\ell(\ell+1)}{2I}, \quad \ell \in \{0, 1, 2, ...\} \tag{46}$$

(b) Based on the claim in part (a), since the hamiltonian is the scalar multiple of the total angular momentum L^2, the corresponding eigenfunctions will be the eigenfunctions of L^2, which are precisely the spherical harmonics Y_ℓ^m.

For each energy level $E_n = \frac{\hbar^2}{2I}n(n+1)$, since the corresponding eigenfunctions have $\ell = n$ (we need $\ell(\ell+1) = n(n+1)$, so with $\ell, n \geq 0$, one needs $\ell = n$), then for the other quantum number m, it's allowed to be $m \in \{-n, -n+1, ..., n-1, \ell\}$, showing there are total of $2n+1$ allowed m, or $2n+1$ linearly independent eigenfunctions. Hence, the degeneracy (i.e. the dimension of the eigenspace) is $2n+1$.

(c) Recall that for a system where angular momentum is conserved, the magnitude of angular momentum $L = Iw$, where I is the system's moment of inertia, w is the angular velocity (or angular frequency).

Then, since the frequency $\nu = \frac{w}{2\pi}$, then for any allowed angular momentum $\hbar\ell$ where $\ell \in \mathbb{N}$, the corresponding angular frequency is $w_\ell = \frac{\hbar\ell}{2\pi}$, so the frequency is $\nu_\ell = \frac{w_\ell}{2\pi} = \frac{\hbar\ell}{2\pi I}$.

(d) By the graph, seems like the gap between each peak / trough is pretty even, and in between 30 to 50 cm^{-1} (also between 50 to 70 cm^{-1}), there are total of 5 gaps, hence one has $\Delta\nu = \frac{50-30}{5} = 4\ \text{cm}^{-1}$ (in the graph).

So, for the actual unit, the difference in frequency is $c\Delta\nu = 4 \cdot 3 \cdot 10^{10} = 1.2 \cdot 10^{11}$ Hz.

Now, given m_1 as mass of ^{12}C, m_2 as mass of ^{16}O, one gets the values as $m_1 \approx 1.993 \cdot 10^{-20}$ kg, and $m_2 \approx 2.656 \cdot 10^{-20}$ kg. One can assume $\Delta\nu = \frac{\hbar}{2\pi I}$ based on part (c), then with $\hbar = 1.055 \cdot 10^{-34}$ J·s, one gets the following:

$$\Delta\nu = \frac{\hbar}{2\pi I} \implies I = \frac{\hbar}{2\pi\Delta\nu} \approx 1.399 \cdot 10^{-46}\ \text{kg} \cdot \text{m}^2 \tag{47}$$

Also, one gets the following:

$$I = \frac{m_1 m_2}{m_1+m_2}a^2 \approx (1.139 \cdot 10^{-20})a^2, \quad a^2 \approx \frac{1.399 \cdot 10^{-46}}{1.139 \cdot 10^{-20}} \approx 1.229 \cdot 10^{-26}\ \text{m}^2 \tag{48}$$

So, one gets the distance between the atoms $a \approx 1.109 \cdot 10^{-13}$m.

$\square$

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Question 5. *Angular Ehrenfest*

(a) Prove the rotational analog to Ehrenfest's theorem: for a particle in a general potential $V(r)$, the rate of change of the expectation value of the orbital angular momentum L is equal to the expectation value of the torque $\mathbf{N} = \mathbf{r} \times (-\nabla V)$,

$$\frac{d}{dt}\langle \mathbf{L} \rangle = \langle \mathbf{N} \rangle$$

(b) Show that $d\langle \mathbf{L} \rangle/dt = 0$ for any spherically symmetric potential. This is the quantum analog of the classical conservation of angular momentum

Proof.

(a) First, using the classical definition, one gets the following:

$$\begin{aligned} \mathbf{N} &= \mathbf{r} \times (-\nabla V) && (49)\\ &= -\left(y\frac{\partial V}{\partial z} - z\frac{\partial V}{\partial y}\right)\hat{x} - \left(z\frac{\partial V}{\partial x} - x\frac{\partial V}{\partial z}\right)\hat{y} - \left(x\frac{\partial V}{\partial y} - y\frac{\partial V}{\partial x}\right)\hat{z} && (50) \end{aligned}$$

Now, assume one can directly promote each individual component to quantum-mechanical operators. Then, in the position basis, the expectation value of torque is givenn as follow:

$$\langle \mathbf{N} \rangle = -\left\langle y\frac{\partial V}{\partial z} - z\frac{\partial V}{\partial y}\right\rangle\hat{x} - \left\langle z\frac{\partial V}{\partial x} - x\frac{\partial V}{\partial z}\right\rangle\hat{y} - \left\langle x\frac{\partial V}{\partial y} - y\frac{\partial V}{\partial x}\right\rangle\hat{z} \tag{51}$$

Now, using the definition $L_x = yp_z - zp_y$, the expectation value is given below (using the position basis, or the wave function):

$$\langle L_x \rangle = \int_{\mathbb{R}^3} \psi^*(yp_z - zp_y)\psi dV \tag{52}$$

Which, incorporating the time derivative, assume the conditions are all satisfied (together with the position and momentum operators being time independent). Using the Schrödinger equation $\frac{\partial \psi}{\partial t} = \frac{(p_x^2+p_y^2+p_z^2)}{i\hbar 2m}\psi + \frac{V}{i\hbar}\psi$, together with the commutation relation $[u, p_v] = i\hbar\delta_{uv}, [p_u, p_v] = 0, [u, v] = 0$ for all $u, v \in \{x, y, z\}$ (which further implies$[u, p_u^2] = 2i\hbar p_u$, and $[u, p_u^3] = 3i\hbar p_u^2$), one gets the following:

$$\begin{aligned} \frac{d\langle L_x \rangle}{dt} &= \frac{d}{dt}\int_{\mathbb{R}^3} \psi^*(yp_z - zp_y)\psi dV = \int_{\mathbb{R}^3} \left(\left(\frac{\partial \psi}{\partial t}\right)^*(yp_z - zp_y)\psi + \psi^*(yp_z - zp_y)\frac{\partial \psi}{\partial t}\right) dV && (53)\\ &= \int_{\mathbb{R}^3} \left(\frac{1}{i\hbar}\left(\frac{p_x^2+p_y^2+p_z^2}{2m} + V\right)\psi\right)^*(yp_z - zp_y)\psi dV + \int_{\mathbb{R}^3} \psi^*(yp_z - zp_y)\frac{1}{i\hbar}\left(\frac{p_x^2+p_y^2+p_z^2}{2m} + V\right)\psi dV && (54)\\ &= \frac{1}{i\hbar}\int_{\mathbb{R}^3} \psi^*\left[yp_z - zp_y, \left(\frac{p_x^2+p_y^2+p_z^2}{2m} + V\right)\right]\psi dV && (55)\\ &= \frac{1}{i\hbar}\left\langle \left[yp_z - zp_y, \left(\frac{p_x^2+p_y^2+p_z^2}{2m} + V\right)\right]\right\rangle && (56) \end{aligned}$$

Which, computing the commutator, one provides the following:

$$\left[yp_z - zp_y, \left(\frac{p_x^2+p_y^2+p_z^2}{2m}+V\right)\right] = \left[yp_z, \frac{p_y^2}{2m}+V\right] - \left[zp_y, \frac{p_z^2}{2m}+V\right] \tag{57}$$
$$= \left(\frac{2i\hbar p_y p_z}{2m} - i\hbar y\frac{\partial V}{\partial z}\right) - \left(\frac{2i\hbar p_y p_z}{2m} - i\hbar z\frac{\partial V}{\partial y}\right) \tag{58}$$
$$= -i\hbar\left(y\frac{\partial V}{\partial z} - z\frac{\partial V}{\partial y}\right) \tag{59}$$

Hence, one gets the following:

$$\frac{d\langle L_x\rangle}{dt} = \frac{1}{i\hbar}\left\langle\left[yp_z - zp_y, \left(\frac{p_x^2+p_y^2+p_z^2}{2m}+V\right)\right]\right\rangle = -\left\langle y\frac{\partial V}{\partial z} - z\frac{\partial V}{\partial y}\right\rangle \tag{60}$$

Do the same calculation, one also gets the following:

$$\frac{d\langle L_y\rangle}{dt} = -\left\langle z\frac{\partial V}{\partial x} - x\frac{\partial V}{\partial z}\right\rangle, \quad \frac{d\langle L_z\rangle}{dt} = -\left\langle x\frac{\partial V}{\partial y} - y\frac{\partial V}{\partial x}\right\rangle \tag{61}$$

Hence, one gets the following for the time derivative of expectation value of angular momentum:

$$\frac{d\langle \mathbf{L}\rangle}{dt} = \frac{d\langle L_x\rangle}{dt}\hat{x} + \frac{d\langle L_y\rangle}{dt}\hat{y} + \frac{d\langle L_z\rangle}{dt}\hat{z} \tag{62}$$
$$= -\left\langle y\frac{\partial V}{\partial z} - z\frac{\partial V}{\partial y}\right\rangle\hat{x} - \left\langle z\frac{\partial V}{\partial x} - x\frac{\partial V}{\partial z}\right\rangle\hat{y} - \left\langle x\frac{\partial V}{\partial y} - y\frac{\partial V}{\partial x}\right\rangle\hat{z} \tag{63}$$
$$= \langle \mathbf{N}\rangle \tag{64}$$

(b) If V is spherically symmetric (i.e. $V = V(r)$ for $r = \sqrt{x^2+y^2+z^2}$), one deduces the following for $r \neq 0$:

$$u \in \{x, y, z\}, \quad \frac{\partial V}{\partial u} = \frac{\partial V}{\partial r}\frac{\partial r}{\partial u} = \frac{u}{r}\frac{\partial V}{\partial r} \tag{65}$$

So, $\nabla V = \sum_{u\in\{x,y,z\}} \frac{u}{r}\frac{\partial V}{\partial r}\hat{u} = \frac{1}{r}\frac{\partial V}{\partial r}\sum_{u\in\{x,y,z\}} u\hat{u} = \frac{1}{r}\frac{\partial V}{\partial r}\mathbf{r}$. Hence, one deduces the following:

$$\mathbf{N} = \mathbf{r} \times (-\nabla V) = -\frac{1}{r}\frac{\partial V}{\partial r}(\mathbf{r}\times\mathbf{r}) = \mathbf{0} \tag{66}$$

So, one also has $\frac{d\langle \mathbf{L}\rangle}{dt} = \langle \mathbf{N}\rangle = \mathbf{0}$.

□